

VLADYSLAV HORBATENKO

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SKILLS

LLM & Agent Orchestration: LangGraph, LangChain, MCP, DSPy, Pydantic, Tool Calling, Multi-Agent Patterns
GenAI Engineering & RAG: Pinecone, ChromaDB, LangSmith (Observability), Guardrails, Evals, Docling
Backend & Infrastructure: Python, Docker, AWS, GCP, CI/CD, REST APIs, PostgreSQL, Redis, Celery
Machine Learning: TensorFlow, PyTorch, Scikit-Learn, Pandas, NumPy, HuggingFace
Other Languages/Tools: Java, MySQL, Django, Next.js, Dokploy, LaTeX

WORK EXPERIENCE

Co-Founder & CTO

PeerMind

Sep 2025 – Present

Denmark, Copenhagen

- Architected and shipped a full-stack AI-powered educational platform (Django, Next.js, PostgreSQL, Redis, Celery) end-to-end, with Test/Production CI/CD pipelines via GitHub Actions and Dokploy, Traefik reverse proxy, and Caddy for secure signed media serving.
- Built a multi-document preprocessing pipeline using VLMs, custom OCR and embedding models for semantic chunking, native document structure analysis, and multi-phase ToC detection, with LLM-based cross-document relationship analysis and merging.
- Implemented hybrid RAG with Pinecone vector search, agentic multi-query retrieval via DSPy, and GraphRAG-style page-span context assembly for high-fidelity content generation.
- Built AI agents for both students and teachers. Multimodal AI Tutor with RAG search, image analysis, and streaming responses, and a ReAct page-editing agent with structured tool use, powered by LangGraph with full LangSmith observability.

AI Engineer

Tryp.com

Sep 2024 – Present

Denmark, Copenhagen

- Developing and optimizing an LLM-based customer support system from scratch: improved multi-turn dialogue handling, achieved 86% intent recognition accuracy, reduced average response time by 83%, and handled 74% of all inbound inquiries ($\approx 10,500$ /month). Also handled monthly conversations to review issues, align on priorities, and improve the system.
- Optimized a website-wide internal LLM system used across Tryp.com within one month of project takeover, handling 50k daily requests while reducing daily usage from 300M to 100M tokens and cutting cost 4x through system-side caching, KV-cache loading, input/context adjustment, and compressed Redis cache payloads.
- Built an agentic travel search agent capable of multi-step reasoning across destinations, pricing, and user constraints.

Co-Founder (Lead AI Engineer)

CoreAI NGO Project

Jan 2025 – April 2026

Denmark, Copenhagen

- Deployed an AI chatbot system based on AWS Bedrock, providing 24/7 multilingual support to 10,000 users within Danish social sector and higher education.
- Built a microservice AWS backend (API Gateway, Lambda, DynamoDB, LightSail) with a RAG pipeline for real-time responses and booking integration (EasyWeek API).
- Integrated Telegram and WhatsApp frontends via API Gateway & Lambda for fast user access.
- Co-authored the AWS Machine Learning Blog post “Build a scalable AI assistant to help refugees using AWS.”

Bachelor Thesis Student

Novo Nordisk

Feb 2025 – Jul 2025

Denmark

- Gained corporate engineering experience by collaborating cross-functionally with research teams to define requirements and align on technical deliverables.
- Built and delivered an end-to-end data extraction tool utilizing a fine-tuned DETR (Detection Transformer) model, achieving 94% precision in locating tables in pharmaceutical PDFs.
- Parsing of complex table structures for tailored data extraction, drastically reducing manual processing time while ensuring high data fidelity.

EDUCATION

Technical University of Denmark DTU: MSc in Human-Centered Artificial Intelligence Sep 2025 - Jun/Jan 2027

Roskilde University RUC: BSc in Computer Science and Mathematics

Sep 2022 - Jun 2025